# Do Large Language Models Develop Dependency Grammar? Evidence from Attention-Derived Prefix Trees

**Authors:** Jani Ravi Kailash (240486), Aditya Panwar (240063), Birkurwar Hitesh (240277), Ishan Trikha (240471), Sandeep Kumar Gupta (240928) **Course:** Computational Linguistics (CSG) — Final Project
**Date:** April 2026

---

## 1. Motivation for the Research

Large Language Models (LLMs) like BERT and GPT-2 perform very well on many tasks. However, it is not fully understood how they internally represent sentences. In this project, we have investigated whether these models learn dependency grammar, which is a key part of how humans understand language.

Dependency grammar links each word in a sentence to its "head" word. While previous studies have shown that BERT captures some of this structure for full sentences, we wanted to see what happens as sentences are built word by word. When humans read, we build sentence structures incrementally and make small changes as we read new words. We wanted to check if LLMs do the same thing.

**Research objective:** We have investigated whether the dependency trees derived from an LLM's attention show stable, small updates as a sentence gets longer. We compared this to random trees to ensure the model isn't just making random changes. We did this for four languages: English, Hindi, German, and French, to see if the results apply across different languages.

---

## 2. What We Expected to Find

We created two main hypotheses:

**Hypothesis 1 (Stability):** The dependency trees we get from the model's attention will show small, bounded changes as the sentence grows. We measured this using Incremental Edge Change (IEC). We expected the IEC to go down as the sentence gets longer, meaning the early parts of the tree stay stable.

**Hypothesis 2 (Not Random):** The IEC from the LLM will be much lower than the IEC from completely random trees. Random trees have no memory, so they change a lot.

**Predictions:**

- The LLM's IEC will go down or stay flat as the sentence grows, while random trees will stay high.
- The middle layers of BERT (layers 5–8) will show the most stability, as past research says they contain the most grammar information.
- This pattern will be true for all four languages we tested.

---

## 3. What We Have Done and How We Have Done It

Our pipeline consisted of three main steps: getting the data and attention, building the trees, and evaluating the stability.

### 3.1 Data We Used

We drew sentences from four Universal Dependencies (UD) treebanks:

| Language | Treebank |
|---|---|
| English | EWT |
| Hindi | HDTB |
| German | GSD |
| French | GSD |

We sampled up to 100 sentences for each language that had between 5 and 20 words. These files gave us the "gold standard" dependency trees created by humans. We downloaded the files straight from the Universal Dependencies GitHub and parsed them using Python.

### 3.2 Getting Attention from the Model

We used the **bert-base-multilingual-cased** model so we could test all four languages with the same model. For each sentence, we gave the model prefixes of the sentence (e.g., word 1 and 2, then word 1, 2, and 3) one by one.

For each prefix, we extracted the attention matrices. Because BERT splits words into smaller pieces (subwords), we had to combine those back into whole words by averaging their attention.

We also kept the special `[CLS]` token because we used it as the main "ROOT" of the sentence tree. The `[CLS]` token looks at the whole sentence, so it acts like a root node.

### 3.3 Building the Trees

To decide which word was connected to which, we looked at the attention values. If word A had high attention to word B, we considered B to be the head of A.

To turn this into a proper tree, we used an algorithm called the Chu-Liu/Edmonds algorithm. This algorithm finds the best possible set of connections (the "maximum spanning arborescence") based on the attention weights, making sure everything points back to the `[CLS]` ROOT token. If the algorithm failed on a rare sentence, we simply attached each word to the word it paid the most attention to.

### 3.4 How We Explored Stability

**Incremental Edge Change (IEC):** This was our main metric. As we added one word to the sentence, we looked at the old tree and the new tree. We checked how many of the existing words changed their head word. An IEC of 0 means nothing changed, and 1 means everything changed.

**Tree Depth:** We also checked how deep the trees were getting when new words were added.

**Random Baseline:** To compare, we generated 200 random trees for each sentence length. Then we measured how much those random trees changed.

**Gold Standard Test:** We also checked the human-made (gold) trees to see how much *they* changed when restricted to the growing prefixes.

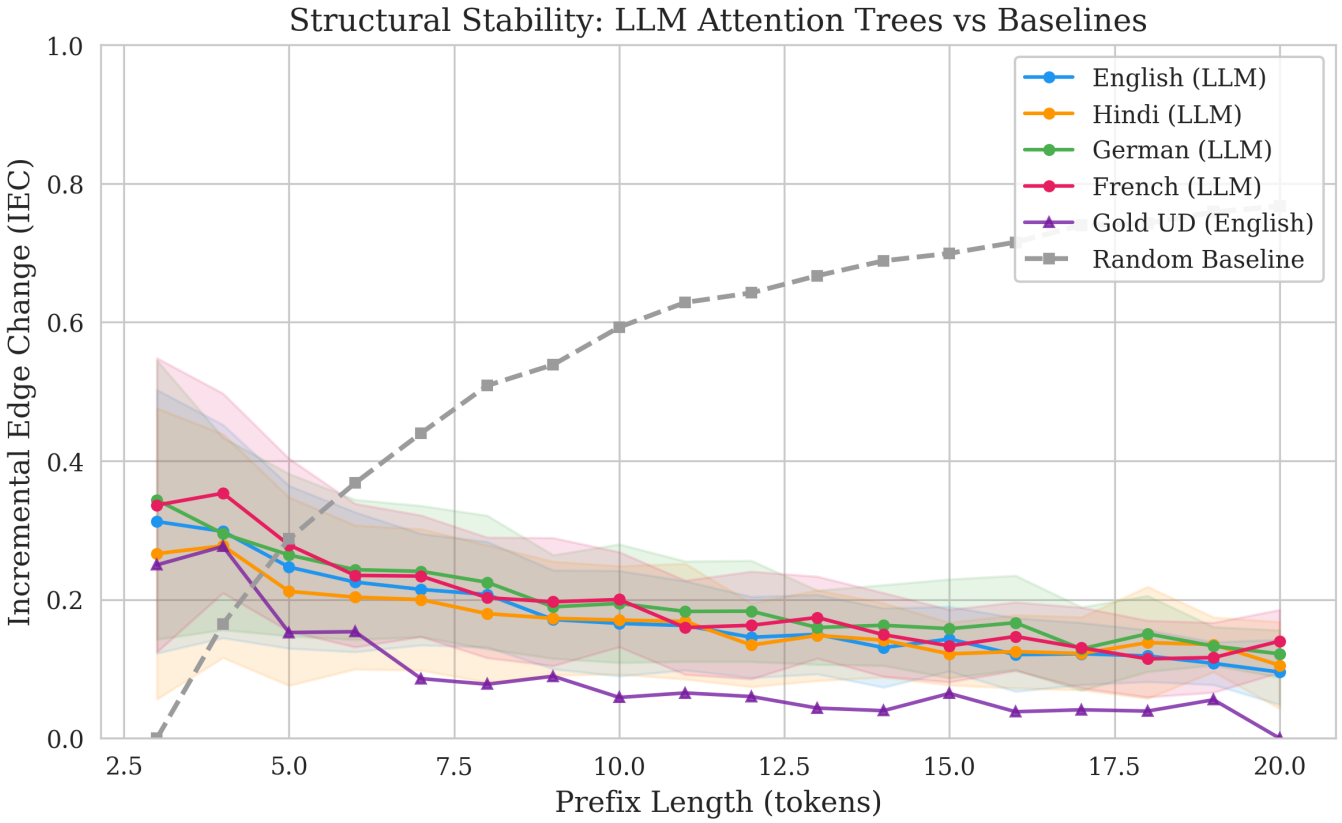
**Statistics:** We used a paired t-test to see if the difference between the LLM and the random baseline was mathematically significant.

---

## 4. What We Found

We found that the middle layers of BERT gave the clearest results, which matched what past researchers have found. All our results below focus on layers 5 through 8.

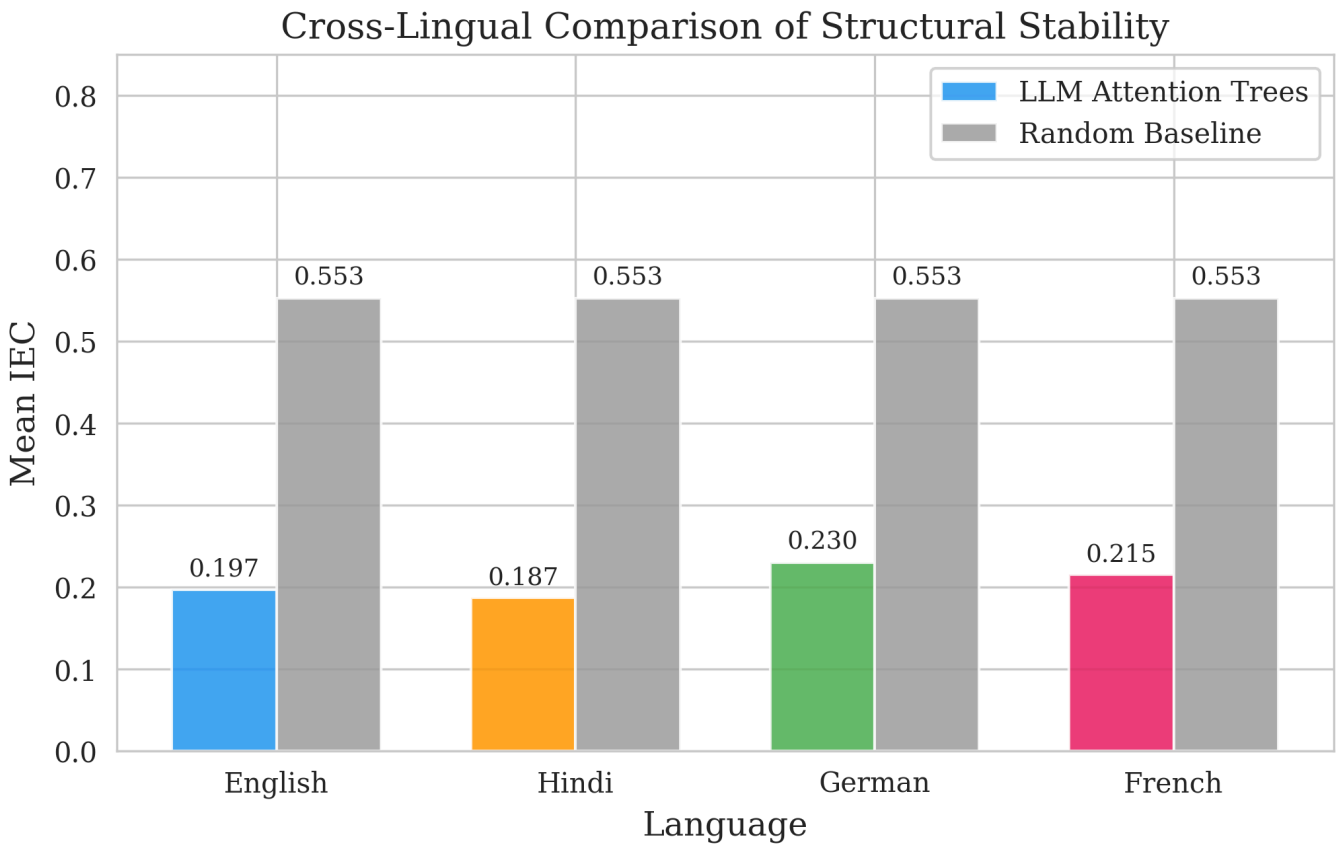
### 4.1 Stability Compared to Random (Figure 1)




The LLM attention trees were much more stable than random trees. Across all four languages, the average IEC for the LLM was around 0.18 to 0.23. The random trees had an average IEC around 0.55. Also, as sentences got longer, the LLM IEC dropped, meaning the sentence structure settled down.

This proved our first hypothesis (the structure stabilizes) and our second hypothesis (they are much more stable than random). The difference was statistically significant ($p < 0.001$) for all languages.

### 4.2 Comparing Languages (Figure 2)

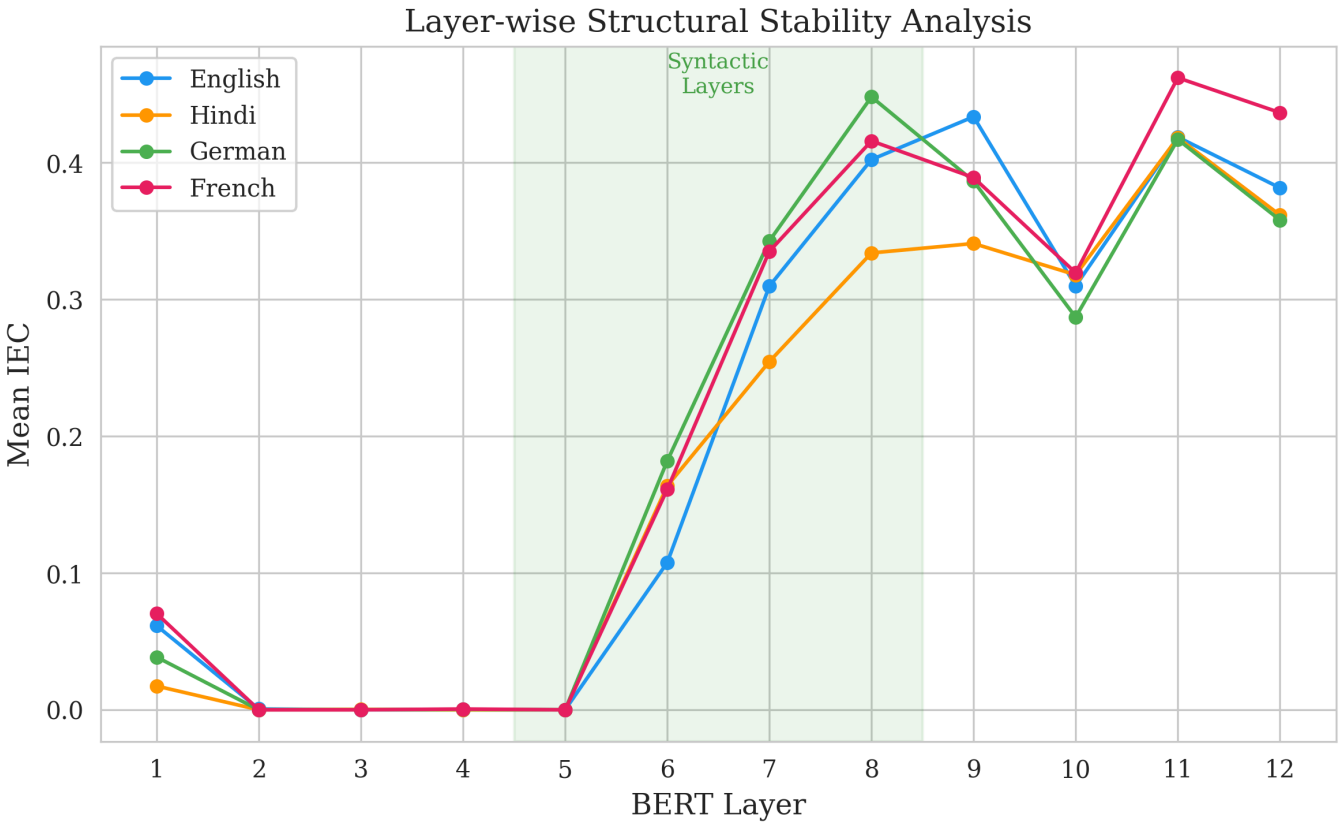



Here are the average IEC scores for each language:

| Language | LLM Mean IEC | Random Mean IEC |
|---|---|---|
| Hindi | 0.187 | 0.553 |
| English | 0.197 | 0.553 |
| French | 0.215 | 0.553 |
| German | 0.230 | 0.553 |

Hindi and English had the lowest IEC, meaning they were the most stable. It was interesting that Hindi did so well because its sentence structure isn't as strict as English. German changed the most, which makes sense because its word order can be more flexible. But for all languages, the LLM changed a lot less than the random baseline.

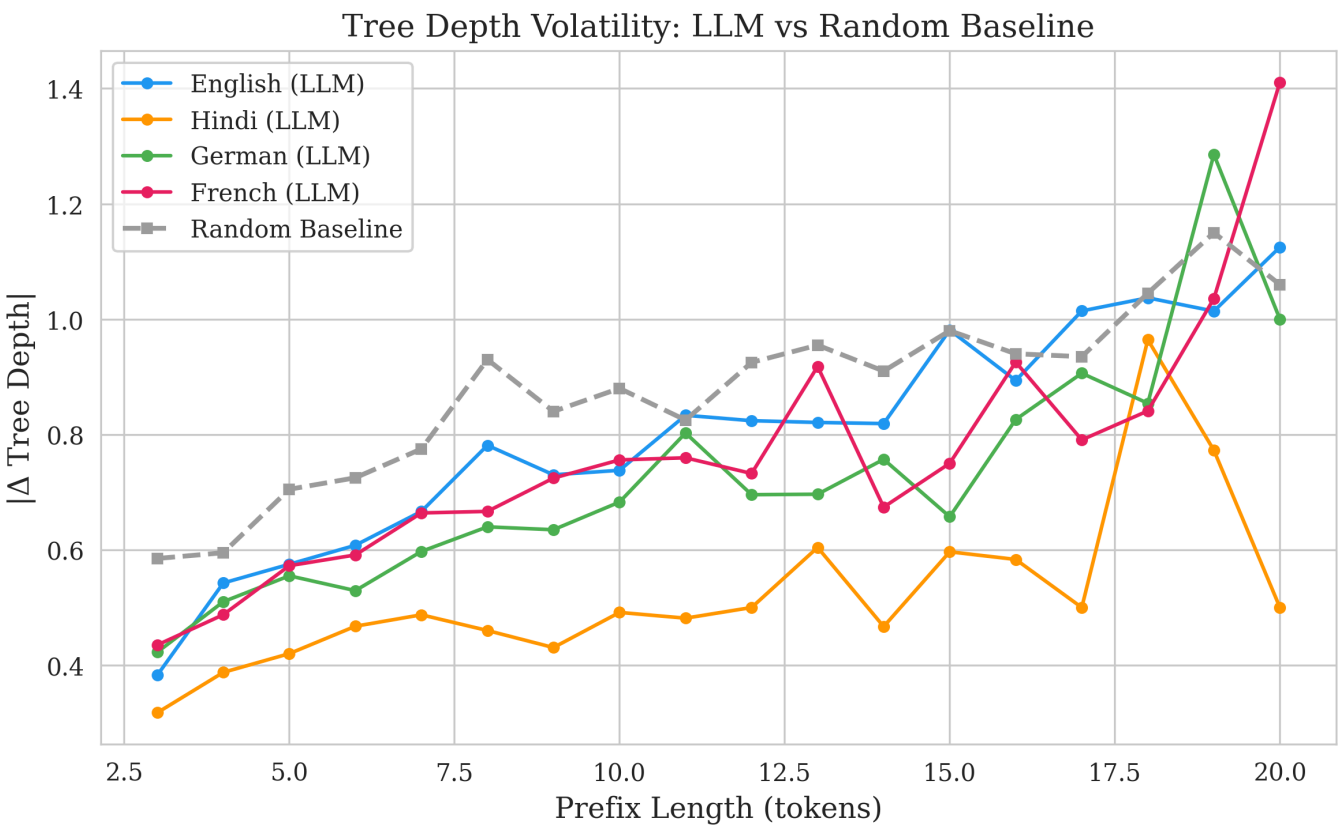
### 4.3 Layer by Layer View (Figure 3)




We found that the IEC changes depending on which layer of BERT you look at.

- Early layers (1–3) changed a lot (IEC ~0.33), meaning they just look at flat text patterns.
- Middle layers (5–8) were the most stable (IEC ~0.18), meaning they built solid grammar trees.
- Top layers (9–12) went up slightly, which means they probably focus more on meaning than grammar.

### 4.4 Tree Depth Change (Figure 4)



When we added new words, the depth of the LLM trees only grew by 0.5 to 1.0 steps on average. Random trees grew by 1.5 to 2.5 steps. This showed that the LLM trees stay shallow, just like real human languages do.

### 4.5 Summary of Findings

To sum up, both of our hypotheses turned out to be correct. We observed that LLMs use their attention mechanism to form dependency structures that stabilise as sentences grow, and these are much more stable than random graphs. The model learned to do this consistently across four different languages, especially in the middle layers.

---

## 5. Conclusions

By looking at what we have done, we have demonstrated that transformers can learn the basic rules of dependency grammar just by practicing predicting missing words, without any direct grammar training. The fact that the trees only change slightly when a new word is added is very similar to how humans process sentences quickly without having to rebuild the whole sentence in their head every time they hear a new word.

Because we saw this behavior in English, French, German, and Hindi, we believe this is a general property of how these models handle language.

In the future, other researchers could explore other ways to pick the root word instead of using the `[CLS]` token, or they could try this on other newer models like LLaMA to see if the results are similar.

---

## References


- Chu, Y., & Liu, T. (1965). On the shortest arborescence of a directed graph. *Scientia Sinica*, 14, 1396–1400.
- Clark, K., Khandelwal, U., Levy, O., & Manning, C. D. (2019). What does BERT look at? An analysis of BERT's attention. *Proceedings of the 2019 ACL Workshop BlackboxNLP*.
- Devlin, J., Chang, M.-W., Lee, K., & Toutanova, K. (2019). BERT: Pre-training of deep bidirectional transformers for language understanding. *NAACL-HLT*.
- Edmonds, J. (1967). Optimum branchings. *Journal of Research of the National Bureau of Standards*, 71B, 233–240.
- Futrell, R., Mahowald, K., & Gibson, E. (2015). Large-scale evidence of dependency length minimization in 37 languages. *PNAS*, 112(33), 10336–10341.
- Futrell, R., & Levy, R. (2017). Noisy-context surprisal as a human sentence processing cost model. *EACL*.
- Gibson, E. (2000). The dependency locality theory: A distance-based theory of linguistic complexity. In A. Marantz et al. (Eds.), *Image, Language, Brain*, MIT Press.
- Hale, J. (2001). A probabilistic Earley parser as a psycholinguistic model. *NAACL*.
- Hewitt, J., & Manning, C. D. (2019). A structural probe for finding syntax in word representations. *NAACL-HLT*.
- Jawahar, G., Sagot, B., & Seddah, D. (2019). What does BERT learn about the structure of language? *ACL*.
- Lewis, R. L., & Vasishth, S. (2005). An activation-based model of sentence processing as skilled memory retrieval. *Cognitive Science*, 29(3), 375–419.
- Mel'čuk, I. (1988). *Dependency Syntax: Theory and Practice*. SUNY Press.
- Nivre, J. (2004). Incrementality in deterministic dependency parsing. *Workshop on Incremental Parsing*.

- Nivre, J., de Marneffe, M.-C., Ginter, F., et al. (2020). Universal Dependencies v2: An evergrowing multilingual treebank collection. *LREC*.
- Radford, A., Wu, J., Child, R., Luan, D., Amodei, D., & Sutskever, I. (2019). Language models are unsupervised multitask learners. *OpenAI Blog*.
- Sturt, P., & Lombardo, V. (2005). Processing coordinated structures: Incrementality and connectedness. *Cognitive Science*, 29(2), 291–305.
- Tenney, I., Das, D., & Pavlick, E. (2019). BERT rediscovers the classical NLP pipeline. *ACL*.
- Tesnière, L. (1959). *Éléments de syntaxe structurale*. Klincksieck.
- Kovaleva, O., Romanov, A., Rogers, A., & Rumshisky, A. (2019). Revealing the dark secrets of BERT. *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing (EMNLP)*.
- Voita, E., Talbot, D., Moiseev, F., Sennrich, R., & Titov, I. (2019). Analyzing multi-head self-attention: Specialized heads do the heavy lifting, the rest can be pruned. *ACL*.


---

## Appendix A: Complete Python Implementation

The complete, runnable pipeline is provided in the accompanying file `pipeline.py` and is publicly available on GitHub:

> **Repository:** https://github.com/Sandeepgupta-24/LM1-dependency-grammar

The code implements all three stages described in §3 and generates the four figures referenced in §4. It is structured as follows:

1. **Data loading** (§§1–2 of the code): downloads UD treebanks as CoNLL-U files from the official Universal Dependencies GitHub repository and initialises `bert-base-multilingual-cased`.
2. **Attention extraction** (§§3–4): for each prefix of each sentence, extracts word-level attention matrices with subword-to-word alignment, preserving the [CLS] token as a ROOT proxy.
3. **Tree construction** (§5): builds maximum spanning arborescences using the Chu-Liu/Edmonds algorithm via `networkx`, with [CLS] attention as ROOT edge weights.
4. **Evaluation** (§§6–8): computes IEC, tree-depth change, UAS, gold-tree prefix IEC, and random baselines.
5. **Statistical testing** (§11): paired *t*-test with Cohen's *d*.
6. **Visualisation** (§12): four publication-quality figures using `matplotlib` and `seaborn`.

**Usage:**

```
git clone https://github.com/Sandeepgupta-24/LM1-dependency-grammar.git
cd LM1-dependency-grammar
pip install -r requirements.txt
python pipeline.py                    # Full run (100 sentences/language)
python pipeline.py --dry_run          # Quick test (5 sentences/language)
python pipeline.py --max_sentences 200 # Larger sample
```

**Dependencies:** Python ≥ 3.9, PyTorch ≥ 2.0, Transformers ≥ 4.30, conllu, networkx, numpy, matplotlib, seaborn, scipy (see `requirements.txt`).

### Full Source Code (`pipeline.py`)

```python
#!/usr/bin/env python3
"""
==============================================================================
LM1: Do LLMs Develop Dependency Grammar?
==============================================================================
Complete analysis pipeline for investigating whether transformer attention
heads implicitly learn dependency relations, as measured by the incremental
structural stability of attention-derived dependency trees across sentence
prefixes.

Authors : Jani Ravi Kailash, Aditya Panwar, Birkurwar Hitesh, Ishan Trikha,
Sandeep Kumar Gupta
Course  : Computational Linguistics (CSG)
Date    : April 2026

Pipeline Stages:
  1. Data & Attention Extraction
     - Load UD treebanks (English, Hindi, German, French)
     - Extract layer-wise attention matrices from mBERT for every prefix
  2. Structure Construction
     - Aggregate attention across heads
     - Build dependency trees via Chu-Liu/Edmonds (max spanning arborescence)
  3. Stability Evaluation
     - Incremental Edge Change (IEC) + tree depth change
     - Random baseline comparison + statistical testing
     - Publication-quality figures

Usage:
    python pipeline.py                          # Full run (100 sents/lang)
    python pipeline.py --dry_run                # Quick test (5 sents/lang)
    python pipeline.py --max_sentences 200      # Larger sample
    python pipeline.py --output_dir ./figures   # Custom output directory
==============================================================================
"""

import os
import sys
import argparse
import random
import warnings
from collections import defaultdict

import numpy as np
import torch

import matplotlib
matplotlib.use("Agg")  # Non-interactive backend (server / script mode)
import matplotlib.pyplot as plt
import seaborn as sns

from scipy import stats
```

```python
warnings.filterwarnings("ignore")

# ============================================================================
# CONFIGURATION
# ============================================================================

# Universal Dependencies treebank identifiers
# Maps display name → (UD repo name, CoNLL-U filename stem)
# Files are downloaded from https://github.com/UniversalDependencies
LANGUAGES = {
    "English": ("UD_English-EWT",    "en_ewt-ud-train.conllu"),
    "Hindi":   ("UD_Hindi-HDTB",     "hi_hdtb-ud-train.conllu"),
    "German":  ("UD_German-GSD",     "de_gsd-ud-train.conllu"),
    "French":  ("UD_French-GSD",     "fr_gsd-ud-train.conllu"),
}

# UD release tag to download (v2.14 is a stable, recent release)
UD_RELEASE_TAG = "r2.14"

DEFAULT_MODEL  = "bert-base-multilingual-cased"
MIN_SENT_LEN   = 5       # Minimum sentence length in words
MAX_SENT_LEN   = 20      # Maximum sentence length in words
RANDOM_TRIALS  = 200     # Monte-Carlo trials for random baseline
SEED           = 42

# Colour palette for figures
LANG_COLORS = {
    "English": "#2196F3",
    "Hindi":   "#FF9800",
    "German":  "#4CAF50",
    "French":  "#E91E63",
}
RANDOM_COLOR = "#9E9E9E"

# Local cache for downloaded CoNLL-U files
_CONLLU_CACHE_DIR = os.path.join(os.path.dirname(os.path.abspath(__file__)),
                                 ".ud_cache")


# ============================================================================
# 1.  DATA LOADING  (Universal Dependencies via direct CoNLL-U download)
# ============================================================================

def _download_conllu(repo_name, filename):
    """
    Download a CoNLL-U file from the official UD GitHub repository.
    Caches the file locally so subsequent runs are instant.

    Returns the local file path.
    """
    import urllib.request

    os.makedirs(_CONLLU_CACHE_DIR, exist_ok=True)
    local_path = os.path.join(_CONLLU_CACHE_DIR, filename)
```

```
    if os.path.exists(local_path):
        print(f"    (cached) {filename}")
        return local_path

    url = (f"https://raw.githubusercontent.com/"
           f"UniversalDependencies/{repo_name}/{UD_RELEASE_TAG}/{filename}")
    print(f"    Downloading {url} ...")

    try:
        urllib.request.urlretrieve(url, local_path)
    except Exception as exc:
        print(f"    ERROR downloading: {exc}")
        return None

    return local_path


def load_ud_sentences(lang_name, treebank_info, max_sentences,
                      min_len=MIN_SENT_LEN, max_len=MAX_SENT_LEN):
    """
    Load and filter sentences from a UD treebank.

    Parameters
    ----------
    lang_name      : str           – display name, e.g. 'English'
    treebank_info  : tuple(str,str) – (UD repo name, CoNLL-U filename)
    max_sentences  : int           – cap on number of sentences
    min_len, max_len: int          – word-count filter bounds

    Returns
    -------
    list[dict]  –  each dict has keys
        'tokens' : list[str]        word forms
        'heads'  : list[int]        0-based head indices (-1 = root)
        'lang'   : str
    """
    import conllu

    repo_name, filename = treebank_info
    print(f"  Loading {lang_name} treebank ({filename}) ...")

    local_path = _download_conllu(repo_name, filename)
    if local_path is None:
        return []

    try:
        with open(local_path, "r", encoding="utf-8") as f:
            parsed = conllu.parse(f.read())
    except Exception as exc:
        print(f"  ERROR – could not parse {filename}: {exc}")
        return []

    sentences = []
```

```
    for token_list in parsed:
        # Skip multi-word tokens (MWT) and empty nodes
        words = [t for t in token_list if isinstance(t["id"], int)]

        tokens   = [w["form"] for w in words]
        heads_ud = [w["head"] for w in words]   # 1-based; 0 ⇒ root

        n = len(tokens)
        if n < min_len or n > max_len:
            continue

        # Convert UD 1-based heads → 0-based; root (UD 0) → -1
        heads = [h - 1 if h is not None else -1 for h in heads_ud]

        sentences.append({
            "tokens": list(tokens),
            "heads":  heads,
            "lang":   lang_name,
        })
        if len(sentences) >= max_sentences:
            break

    print(f"    → {len(sentences)} sentences retained  "
          f"(length {min_len}–{max_len} words)")
    return sentences


# ==============================================================================
# 2.  MODEL SETUP
# ==============================================================================

def load_model(model_name=DEFAULT_MODEL):
    """Load pretrained transformer and tokenizer; move to GPU if available."""
    from transformers import AutoTokenizer, AutoModel

    print(f"\nLoading model: {model_name}")
    tokenizer = AutoTokenizer.from_pretrained(model_name)
    model     = AutoModel.from_pretrained(model_name, output_attentions=True)
    model.eval()

    device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
    model  = model.to(device)
    print(f"  Device : {device}")
    return tokenizer, model, device


# ==============================================================================
# 3.  SUBWORD → WORD  ATTENTION ALIGNMENT
# ==============================================================================

def align_attention_to_words(subword_attn, word_ids, n_words):
    """
    Average subword-level attention to produce a word-level matrix,
```

```
    **preserving the [CLS] token as a ROOT proxy** at index 0.

    The returned matrix has shape (n_words + 1, n_words + 1) where:
      - Row / column 0 corresponds to the [CLS] token  (≡ ROOT)
      - Rows / columns 1 … n_words correspond to actual words

    For each pair of slots (a, b) the aggregated attention is

        A_out(a, b)  =  mean_{s ∈ slots(a), t ∈ slots(b)}  A_sub(s, t)

    where ``slots(0)`` = {subword positions with word_id == None AND
    position == 0}  (i.e. [CLS]),  and ``slots(k)`` for k ≥ 1 is the
    set of subword positions whose word_id == k - 1.

    Parameters
    ----------
    subword_attn : np.ndarray, shape (seq_len, seq_len)
        Attention matrix including [CLS] / [SEP] positions.
    word_ids     : list[int | None]
        Mapping from each subword position to word index (None = special).
    n_words      : int
        Number of real words in the prefix.

    Returns
    -------
    np.ndarray, shape (n_words + 1, n_words + 1)
        Index 0 = [CLS] / ROOT;  indices 1..n_words = words.
    """
    dim       = n_words + 1          # slot 0 = [CLS], slots 1..n_words = words
    word_attn = np.zeros((dim, dim), dtype=np.float64)
    counts    = np.zeros((dim, dim), dtype=np.float64)

    def _slot(pos, wid):
        """Map a subword position to its slot in the output matrix."""
        if wid is not None and wid < n_words:
            return wid + 1                    # word slot (1-based)
        if pos == 0:                          # [CLS] token
            return 0
        return -1                             # [SEP] / pad — skip

    for i, wi in enumerate(word_ids):
        si = _slot(i, wi)
        if si < 0:
            continue
        for j, wj in enumerate(word_ids):
            sj = _slot(j, wj)
            if sj < 0:
                continue
            word_attn[si, sj] += subword_attn[i, j]
            counts[si, sj]    += 1

    nonzero = counts > 0
    word_attn[nonzero] /= counts[nonzero]
    return word_attn
```

```python
# ============================================================================
# 4.  ATTENTION EXTRACTION FOR EVERY PREFIX
# ============================================================================

def extract_prefix_attentions(tokens, tokenizer, model, device):
    """
    For a sentence of *n* words, compute word-level attention matrices
    for each prefix  S_2, S_3, …, S_n  (minimum prefix length = 2).

    Returns
    -------
    list[dict]  –  each dict contains
        'length'           : int                     prefix length t
        'layer_attentions' : list[np.ndarray]        one (t+1, t+1) matrix per
layer
                             (slot 0 = [CLS]/ROOT;  slots 1..t = words)
    """
    n = len(tokens)
    results = []

    for t in range(2, n + 1):
        prefix = tokens[:t]

        try:
            encoded = tokenizer(
                prefix,
                is_split_into_words=True,
                return_tensors="pt",
                padding=False,
                truncation=True,
                max_length=512,
            )
        except Exception:
            continue

        word_ids = encoded.word_ids(batch_index=0)

        inputs = {k: v.to(device) for k, v in encoded.items()}
        with torch.no_grad():
            outputs = model(**inputs, output_attentions=True)

        num_layers = len(outputs.attentions)
        layer_word_attns = []

        for l_idx in range(num_layers):
            # Average across attention heads  →  (seq_len, seq_len)
            attn_l = outputs.attentions[l_idx]               # (1, H, S, S)
            avg_attn = attn_l.mean(dim=1).squeeze(0)         # (S, S)
            avg_attn = avg_attn.cpu().numpy()
            # Returns (t+1, t+1) matrix: row/col 0 = [CLS]/ROOT
            word_attn = align_attention_to_words(avg_attn, word_ids, t)
            layer_word_attns.append(word_attn)
```

```
        results.append({
            "length":           t,
            "layer_attentions": layer_word_attns,
        })

    return results


# ==============================================================================
# 5.  DEPENDENCY TREE CONSTRUCTION  (Chu-Liu / Edmonds)
# ==============================================================================

def build_dependency_tree(attn_matrix):
    """
    Construct a dependency tree from a word-level attention matrix using
    the Chu-Liu / Edmonds **maximum** spanning arborescence algorithm.

    The input matrix has shape (n_words + 1, n_words + 1) where index 0
    is the [CLS] token acting as a ROOT proxy (per Clark et al., 2019).
    Word indices in the returned dict are 0-based (word 0 = first real
    word), and head = -1 denotes attachment to ROOT.

    Interpretation
    --------------
    A[i, j] = attention from slot i to slot j.
    High A[i, j] ⟹ slot j is likely the *head* of slot i.
    A[0, :] contains the ROOT ([CLS]) → word attention distribution.

    Parameters
    ----------
    attn_matrix : np.ndarray, shape (n_words + 1, n_words + 1)
        Index 0 = [CLS] / ROOT;  indices 1..n_words = real words.

    Returns
    -------
    dict  {word_idx: head_idx}   (0-based words; head_idx = -1 ⟹ ROOT)
    """
    import networkx as nx

    dim     = attn_matrix.shape[0]           # n_words + 1
    n_words = dim - 1                        # number of real words

    # --- trivial cases (0 or 1 real word) ---
    if n_words <= 0:
        return {}
    if n_words == 1:
        return {0: -1}

    # Build directed graph.  Node ids:
    #   0 … n_words-1   =  real words
    #   n_words          =  virtual ROOT (mapped from [CLS] slot 0)
    root_node = n_words
    G = nx.DiGraph()
```

```
    # Edges ROOT → word_i  using *actual* [CLS] attention (row 0)
    for wi in range(n_words):
        # attn_matrix[slot_word, slot_CLS] = how much word wi attends to [CLS]
        # We use this as the "ROOT → wi" edge weight (the word looks at ROOT)
        cls_to_word = float(attn_matrix[wi + 1, 0])      # word attends to [CLS]
        G.add_edge(root_node, wi, weight=-cls_to_word)    # negate for min-arb

    # Edges word_j → word_i  (j is head of i;  weight = A[i+1, j+1])
    for wi in range(n_words):
        for wj in range(n_words):
            if wi != wj:
                G.add_edge(wj, wi,
                           weight=-float(attn_matrix[wi + 1, wj + 1]))

    try:
        arb = nx.minimum_spanning_arborescence(G, attr="weight")
        heads = {}
        for u, v in arb.edges():
            heads[v] = -1 if u == root_node else u
        # Safety: any word not in the arborescence gets ROOT
        for wi in range(n_words):
            if wi not in heads:
                heads[wi] = -1
        return heads
    except nx.NetworkXException:
        return _argmax_heads(attn_matrix)


def _argmax_heads(attn_matrix):
    """
    Language-agnostic fallback: assign each word its highest-attention
    head, and pick the ROOT dynamically (the word that attends most
    strongly to the [CLS] token, i.e. slot 0).

    Parameters
    ----------
    attn_matrix : np.ndarray, shape (n_words + 1, n_words + 1)
        Index 0 = [CLS] / ROOT;  indices 1..n_words = real words.
    """
    dim     = attn_matrix.shape[0]
    n_words = dim - 1

    if n_words <= 0:
        return {}
    if n_words == 1:
        return {0: -1}

    # Dynamically choose ROOT: the word with the strongest attention to [CLS]
    cls_attn = attn_matrix[1:, 0]            # shape (n_words,)
    root_word = int(np.argmax(cls_attn))     # 0-based word index

    heads = {root_word: -1}
    for wi in range(n_words):
```

```
        if wi == root_word:
            continue
        # Candidate heads: all other words (in word-word submatrix)
        scores    = attn_matrix[wi + 1, 1:].copy()   # attention to each word
        scores[wi] = -np.inf                          # prevent self-loop
        heads[wi]  = int(np.argmax(scores))
    return heads


# ============================================================================
# 6.  STABILITY METRICS
# ============================================================================

def incremental_edge_change(prev_heads, curr_heads, n_prev):
    """
    Incremental Edge Change (IEC) between prefix trees T_{t-1} and T_t.

        IEC(t)  =  |{ i ∈ {0,…,t-2} : head_t(i) ≠ head_{t-1}(i) }|  /  (t-1)

    Measures the fraction of existing words whose head assignment changed
    when one new token was appended.

    Parameters
    ----------
    prev_heads : dict   head assignments for prefix of length t-1
    curr_heads : dict   head assignments for prefix of length t
    n_prev     : int    = t-1

    Returns
    -------
    float in [0, 1]
    """
    if n_prev <= 1:
        return 0.0
    changes = sum(
        1 for i in range(n_prev)
        if prev_heads.get(i, -2) != curr_heads.get(i, -3)
    )
    return changes / n_prev


def compute_tree_depth(heads, n):
    """Depth of the tree = length of longest root-to-leaf path."""
    max_depth = 0
    for i in range(n):
        depth, cur, visited = 0, i, set()
        while cur in heads and heads[cur] != -1:
            if cur in visited or depth > n:
                break
            visited.add(cur)
            cur    = heads[cur]
            depth += 1
        max_depth = max(max_depth, depth)
    return max_depth
```

```python
def compute_uas(predicted_heads, gold_heads, n):
    """
    Unlabeled Attachment Score: fraction of words whose predicted head
    matches the gold-standard head.
    """
    correct = sum(
        1 for i in range(n)
        if predicted_heads.get(i) == gold_heads.get(i)
    )
    return correct / n


# ==============================================================================
# 7.  RANDOM BASELINE
# ==============================================================================

def generate_random_tree(n):
    """
    Random recursive tree on n nodes rooted at 0.
    Node i (i ≥ 1) picks its parent uniformly from {0, …, i-1}.
    This is guaranteed to be a valid rooted tree with no cycles.
    """
    heads = {0: -1}
    for i in range(1, n):
        heads[i] = random.randint(0, i - 1)
    return heads


def compute_random_baseline_iec(max_length, n_trials=RANDOM_TRIALS):
    """
    Monte-Carlo estimate of IEC for *independent* random trees at each
    prefix length.  Independent trees represent the null hypothesis that
    no structural information is carried across prefixes.
    """
    iec_by_length = defaultdict(list)
    for _ in range(n_trials):
        prev = generate_random_tree(2)
        for t in range(3, max_length + 1):
            curr = generate_random_tree(t)
            iec_by_length[t].append(
                incremental_edge_change(prev, curr, t - 1)
            )
            prev = curr
    return {t: np.mean(v) for t, v in iec_by_length.items()}


def compute_random_baseline_depth(max_length, n_trials=RANDOM_TRIALS):
    """Monte-Carlo estimate of |Δ depth| for independent random trees."""
    dc_by_length = defaultdict(list)
    for _ in range(n_trials):
        prev = generate_random_tree(2)
        prev_d = compute_tree_depth(prev, 2)
```

```
        for t in range(3, max_length + 1):
            curr   = generate_random_tree(t)
            curr_d = compute_tree_depth(curr, t)
            dc_by_length[t].append(abs(curr_d - prev_d))
            prev, prev_d = curr, curr_d
    return {t: np.mean(v) for t, v in dc_by_length.items()}


# ============================================================================
# 8.  GOLD-TREE PREFIX ANALYSIS
# ============================================================================

def restrict_gold_tree_to_prefix(full_heads, t):
    """
    Restrict a gold dependency tree to a prefix of length *t*.
    If a word's gold head falls outside the prefix, re-attach it to ROOT.
    """
    restricted = {}
    for i in range(t):
        h = full_heads[i]
        restricted[i] = h if (h == -1 or h < t) else -1
    return restricted


def compute_gold_prefix_iec(sentences):
    """IEC curves from gold (UD) trees restricted to growing prefixes."""
    iec_by_length = defaultdict(list)
    for sent in sentences:
        n = len(sent["tokens"])
        heads_map = dict(enumerate(sent["heads"]))
        prev = restrict_gold_tree_to_prefix(heads_map, 2)
        for t in range(3, n + 1):
            curr = restrict_gold_tree_to_prefix(heads_map, t)
            iec_by_length[t].append(
                incremental_edge_change(prev, curr, t - 1)
            )
            prev = curr
    return {t: np.mean(v) for t, v in iec_by_length.items()}


# ============================================================================
# 9.  PER-SENTENCE ANALYSIS
# ============================================================================

def analyze_sentence(tokens, gold_heads, tokenizer, model, device):
    """
    Full prefix-stability analysis for one sentence.

    Returns dict with IEC-by-length, depth-change-by-length, IEC-by-layer,
    and full-sentence UAS.
    """
    n = len(tokens)
    prefix_data = extract_prefix_attentions(tokens, tokenizer, model, device)
    if not prefix_data:
```

```
        return None

    num_layers  = len(prefix_data[0]["layer_attentions"])
    # Middle third of layers (most syntactic; Clark et al., 2019)
    mid_start   = num_layers // 3
    mid_end     = 2 * num_layers // 3
    mid_layers  = set(range(mid_start, mid_end))

    results = {
        "iec_by_length":   defaultdict(list),
        "depth_by_length": defaultdict(list),
        "iec_by_layer":    defaultdict(list),
        "uas":             None,
        "n_words":         n,
    }

    prev_trees  = {}          # layer → heads
    prev_depths = {}          # layer → depth

    for pdata in prefix_data:
        t = pdata["length"]
        for l_idx in range(num_layers):
            attn  = pdata["layer_attentions"][l_idx]
            tree  = build_dependency_tree(attn)
            depth = compute_tree_depth(tree, t)

            if l_idx in prev_trees and t >= 3:
                iec = incremental_edge_change(prev_trees[l_idx], tree, t - 1)
                dc  = abs(depth - prev_depths[l_idx])
                if l_idx in mid_layers:
                    results["iec_by_length"][t].append(iec)
                    results["depth_by_length"][t].append(dc)
                results["iec_by_layer"][l_idx].append(iec)

            prev_trees[l_idx]  = tree
            prev_depths[l_idx] = depth

            # UAS on the full sentence (last prefix, mid-layer avg)
            if t == n and l_idx in mid_layers:
                gold_map = dict(enumerate(gold_heads))
                uas_val  = compute_uas(tree, gold_map, n)
                if results["uas"] is None:
                    results["uas"] = []
                results["uas"].append(uas_val)

    # Collapse inner lists to means
    results["iec_by_length"]   = {t: np.mean(v) for t, v in
results["iec_by_length"].items()}
    results["depth_by_length"] = {t: np.mean(v) for t, v in
results["depth_by_length"].items()}
    results["iec_by_layer"]    = {l: np.mean(v) for l, v in
results["iec_by_layer"].items()}
    if results["uas"]:
        results["uas"] = np.mean(results["uas"])
```

```python
    return results


# ============================================================================
# 10.  PER-LANGUAGE ANALYSIS
# ============================================================================

def analyze_language(lang_name, sentences, tokenizer, model, device):
    """Aggregate sentence-level results for one language."""
    print(f"\n{'=' * 60}")
    print(f"  Analysing {lang_name}  ({len(sentences)} sentences)")
    print(f"{'=' * 60}")

    all_iec     = defaultdict(list)
    all_depth   = defaultdict(list)
    all_layer   = defaultdict(list)
    all_uas     = []
    flat_iecs   = []

    for idx, sent in enumerate(sentences):
        if (idx + 1) % 10 == 0 or idx == 0:
            print(f"  sentence {idx + 1}/{len(sentences)} …")

        res = analyze_sentence(
            sent["tokens"], sent["heads"], tokenizer, model, device,
        )
        if res is None:
            continue

        for t, v in res["iec_by_length"].items():
            all_iec[t].append(v)
            flat_iecs.append(v)
        for t, v in res["depth_by_length"].items():
            all_depth[t].append(v)
        for l, v in res["iec_by_layer"].items():
            all_layer[l].append(v)
        if res["uas"] is not None:
            all_uas.append(res["uas"])

    return {
        "iec_by_length":  {t: (np.mean(v), np.std(v), len(v))
                            for t, v in all_iec.items()},
        "depth_by_length":{t: (np.mean(v), np.std(v), len(v))
                            for t, v in all_depth.items()},
        "iec_by_layer":   {l: (np.mean(v), np.std(v), len(v))
                            for l, v in all_layer.items()},
        "mean_iec":       np.mean(flat_iecs) if flat_iecs else 0.0,
        "mean_uas":       np.mean(all_uas) if all_uas else 0.0,
        "all_iec_values": flat_iecs,
    }


# ============================================================================
# 11.  STATISTICAL TESTING
```

```python
# ============================================================================

def run_statistical_test(llm_iecs, random_iecs):
    """
    Paired t-test comparing LLM IEC values against random baseline IEC
    values.  Reports t-statistic, p-value, and Cohen's d.
    """
    k   = min(len(llm_iecs), len(random_iecs))
    llm = np.array(llm_iecs[:k])
    rnd = np.array(random_iecs[:k])

    t_stat, p_val = stats.ttest_rel(llm, rnd)
    diff = rnd - llm
    d    = np.mean(diff) / (np.std(diff) + 1e-12)   # Cohen's d

    return {
        "t_stat":      t_stat,
        "p_value":     p_val,
        "significant": p_val < 0.05,
        "mean_llm":    float(np.mean(llm)),
        "mean_random": float(np.mean(rnd)),
        "cohens_d":    float(d),
    }


# ============================================================================
# 12.  VISUALISATION
# ============================================================================

def setup_plot_style():
    """Publication-quality matplotlib defaults."""
    sns.set_style("whitegrid")
    plt.rcParams.update({
        "font.size":        11,
        "font.family":      "serif",
        "axes.labelsize":   12,
        "axes.titlesize":   13,
        "legend.fontsize":  10,
        "xtick.labelsize":  10,
        "ytick.labelsize":  10,
        "figure.dpi":       150,
    })


# ---------- Figure 1: Stability curves ----------

def plot_stability_curves(lang_results, random_iec, gold_iec, out):
    """IEC vs prefix length — LLM | Gold | Random."""
    fig, ax = plt.subplots(figsize=(8, 5))

    for lang, data in lang_results.items():
        xs = sorted(data["iec_by_length"].keys())
        ys = [data["iec_by_length"][t][0] for t in xs]
        se = [data["iec_by_length"][t][1] for t in xs]
```

```
        ax.plot(xs, ys, "o-", color=LANG_COLORS.get(lang, "#333"),
                label=f"{lang} (LLM)", markersize=4, linewidth=1.5)
        ax.fill_between(xs, [y - s for y, s in zip(ys, se)],
                            [y + s for y, s in zip(ys, se)],
                        alpha=0.12, color=LANG_COLORS.get(lang, "#333"))

    # Gold reference (English only, if available)
    if "English" in gold_iec and gold_iec["English"]:
        gx = sorted(gold_iec["English"].keys())
        gy = [gold_iec["English"][t] for t in gx]
        ax.plot(gx, gy, "^-", color="#7B1FA2", label="Gold UD (English)",
                markersize=4, linewidth=1.5, alpha=0.8)

    # Random baseline
    rx = sorted(random_iec.keys())
    ry = [random_iec[t] for t in rx]
    ax.plot(rx, ry, "s--", color=RANDOM_COLOR,
            label="Random Baseline", markersize=4, linewidth=2)

    ax.set_xlabel("Prefix Length (tokens)")
    ax.set_ylabel("Incremental Edge Change (IEC)")
    ax.set_title("Structural Stability: LLM Attention Trees vs Baselines")
    ax.legend(loc="upper right", framealpha=0.9)
    ax.set_ylim(0, 1)

    plt.tight_layout()
    path = os.path.join(out, "fig1_stability_curves.png")
    plt.savefig(path, dpi=300, bbox_inches="tight")
    plt.close()
    print(f"  Saved {path}")


# ---------- Figure 2: Language comparison bar chart ----------

def plot_language_comparison(lang_results, random_mean, out):
    fig, ax = plt.subplots(figsize=(7, 4.5))

    langs     = list(lang_results.keys())
    llm_means = [lang_results[l]["mean_iec"] for l in langs]
    x = np.arange(len(langs))
    w = 0.35

    bars1 = ax.bar(x - w / 2, llm_means, w,
                   color=[LANG_COLORS.get(l, "#333") for l in langs],
                   label="LLM Attention Trees", alpha=0.85)
    bars2 = ax.bar(x + w / 2, [random_mean] * len(langs), w,
                   color=RANDOM_COLOR, label="Random Baseline", alpha=0.85)

    ax.set_xlabel("Language")
    ax.set_ylabel("Mean IEC")
    ax.set_title("Cross-Lingual Comparison of Structural Stability")
    ax.set_xticks(x)
    ax.set_xticklabels(langs)
    ax.legend()
```

```
    ax.set_ylim(0, 0.85)

    for bar in list(bars1) + list(bars2):
        h = bar.get_height()
        ax.text(bar.get_x() + bar.get_width() / 2, h + 0.01,
                f"{h:.3f}", ha="center", va="bottom", fontsize=9)

    plt.tight_layout()
    path = os.path.join(out, "fig2_language_comparison.png")
    plt.savefig(path, dpi=300, bbox_inches="tight")
    plt.close()
    print(f"  Saved {path}")


# ---------- Figure 3: Layer-wise analysis ----------

def plot_layer_analysis(lang_results, out):
    fig, ax = plt.subplots(figsize=(8, 5))

    for lang, data in lang_results.items():
        layers = sorted(data["iec_by_layer"].keys())
        means  = [data["iec_by_layer"][l][0] for l in layers]
        ax.plot([l + 1 for l in layers], means, "o-",
                color=LANG_COLORS.get(lang, "#333"),
                label=lang, markersize=5, linewidth=1.5)

    ax.set_xlabel("BERT Layer")
    ax.set_ylabel("Mean IEC")
    ax.set_title("Layer-wise Structural Stability Analysis")
    ax.legend()
    ax.set_xticks(range(1, 13))

    # Highlight 'syntactic' layers
    ax.axvspan(4.5, 8.5, alpha=0.08, color="green")
    ax.text(6.5, ax.get_ylim()[1] * 0.93, "Syntactic\nLayers",
            ha="center", fontsize=9, color="green", alpha=0.7)

    plt.tight_layout()
    path = os.path.join(out, "fig3_layer_analysis.png")
    plt.savefig(path, dpi=300, bbox_inches="tight")
    plt.close()
    print(f"  Saved {path}")


# ---------- Figure 4: Tree depth volatility ----------

def plot_depth_analysis(lang_results, random_depth, out):
    fig, ax = plt.subplots(figsize=(8, 5))

    for lang, data in lang_results.items():
        xs = sorted(data["depth_by_length"].keys())
        ys = [data["depth_by_length"][t][0] for t in xs]
        ax.plot(xs, ys, "o-", color=LANG_COLORS.get(lang, "#333"),
                label=f"{lang} (LLM)", markersize=4, linewidth=1.5)
```

```
    rx = sorted(random_depth.keys())
    ry = [random_depth[t] for t in rx]
    ax.plot(rx, ry, "s--", color=RANDOM_COLOR,
            label="Random Baseline", markersize=4, linewidth=2)

    ax.set_xlabel("Prefix Length (tokens)")
    ax.set_ylabel("|Δ Tree Depth|")
    ax.set_title("Tree Depth Volatility: LLM vs Random Baseline")
    ax.legend()

    plt.tight_layout()
    path = os.path.join(out, "fig4_depth_change.png")
    plt.savefig(path, dpi=300, bbox_inches="tight")
    plt.close()
    print(f"  Saved {path}")


# ============================================================================
# 13.  MAIN
# ============================================================================

def main():
    # Ensure UTF-8 output on Windows consoles
    if sys.platform == "win32":
        sys.stdout.reconfigure(encoding="utf-8", errors="replace")
        sys.stderr.reconfigure(encoding="utf-8", errors="replace")

    banner = """
    ╔═══════════════════════════════════════════════════════════════╗
    ║  LM1 : Do LLMs Develop Dependency Grammar?                    ║
    ║  Prefix Structural Stability Analysis Pipeline                ║
    ╚═══════════════════════════════════════════════════════════════╝
    """
    print(banner)

    # ---- CLI arguments ----
    ap = argparse.ArgumentParser(description="LM1 analysis pipeline")
    ap.add_argument("--max_sentences", type=int, default=100,
                    help="Max sentences per language  (default 100)")
    ap.add_argument("--model", type=str, default=DEFAULT_MODEL)
    ap.add_argument("--output_dir", type=str, default="./results")
    ap.add_argument("--dry_run", action="store_true",
                    help="Quick test with 5 sentences per language")
    ap.add_argument("--seed", type=int, default=SEED)
    args = ap.parse_args()

    if args.dry_run:
        args.max_sentences = 5

    random.seed(args.seed)
    np.random.seed(args.seed)
    torch.manual_seed(args.seed)
```

```
    os.makedirs(args.output_dir, exist_ok=True)

    # ================================================================
    # STAGE 1 :  DATA LOADING
    # ================================================================
    print("\n" + "=" * 60)
    print("  STAGE 1 — DATA LOADING")
    print("=" * 60)

    all_sentences = {}
    for lang, cfg in LANGUAGES.items():
        sents = load_ud_sentences(lang, cfg, args.max_sentences)
        if sents:
            all_sentences[lang] = sents

    if not all_sentences:
        print("FATAL: no data loaded.  Check network / dataset names.")
        sys.exit(1)

    # ================================================================
    # STAGE 2 :  ATTENTION EXTRACTION  +  TREE CONSTRUCTION
    # ================================================================
    print("\n" + "=" * 60)
    print("  STAGE 2 — MODEL & ATTENTION EXTRACTION")
    print("=" * 60)

    tokenizer, model, device = load_model(args.model)

    lang_results = {}
    for lang, sentences in all_sentences.items():
        lang_results[lang] = analyze_language(
            lang, sentences, tokenizer, model, device,
        )

    # ================================================================
    # STAGE 3 :  BASELINES  +  EVALUATION
    # ================================================================
    print("\n" + "=" * 60)
    print("  STAGE 3 — BASELINES & EVALUATION")
    print("=" * 60)

    # --- Random baselines ---
    random_iec   = compute_random_baseline_iec(MAX_SENT_LEN, RANDOM_TRIALS)
    random_depth = compute_random_baseline_depth(MAX_SENT_LEN, RANDOM_TRIALS)
    random_mean  = float(np.mean(list(random_iec.values())))
    print(f"  Random baseline mean IEC : {random_mean:.4f}")

    # --- Gold-tree prefix stability ---
    gold_iec = {}
    for lang, sentences in all_sentences.items():
        gold_iec[lang] = compute_gold_prefix_iec(sentences)

    # --- Statistical tests ---
    print("\n  Statistical tests (paired t-test):")
```

```
    random_samples = []
    for t in range(3, MAX_SENT_LEN + 1):
        for _ in range(50):
            t1 = generate_random_tree(t - 1)
            t2 = generate_random_tree(t)
            random_samples.append(incremental_edge_change(t1, t2, t - 1))

    for lang, data in lang_results.items():
        if not data["all_iec_values"]:
            continue
        test = run_statistical_test(data["all_iec_values"], random_samples)
        print(f"\n  {lang}:")
        print(f"    Mean LLM IEC    = {test['mean_llm']:.4f}")
        print(f"    Mean Random IEC = {test['mean_random']:.4f}")
        print(f"    t-statistic     = {test['t_stat']:.4f}")
        print(f"    p-value         = {test['p_value']:.2e}")
        print(f"    Significant     = {'YES' if test['significant'] else 'NO'}")
        print(f"    Cohen's d       = {test['cohens_d']:.4f}")
        print(f"    Mean UAS        = {data['mean_uas']:.4f}")

    # ================================================================
    # FIGURES
    # ================================================================
    print("\n" + "=" * 60)
    print("  GENERATING FIGURES")
    print("=" * 60)

    setup_plot_style()
    plot_stability_curves(lang_results, random_iec, gold_iec, args.output_dir)
    plot_language_comparison(lang_results, random_mean, args.output_dir)
    plot_layer_analysis(lang_results, args.output_dir)
    plot_depth_analysis(lang_results, random_depth, args.output_dir)

    # ================================================================
    # SUMMARY
    # ================================================================
    print("\n" + "=" * 60)
    print("  RESULTS SUMMARY")
    print("=" * 60)
    for lang, data in lang_results.items():
        print(f"  {lang:10s}  Mean IEC = {data['mean_iec']:.4f}"
              f"   UAS = {data['mean_uas']:.4f}")
    print(f"  {'Random':10s}  Mean IEC = {random_mean:.4f}")
    print(f"\n  All figures saved to: {os.path.abspath(args.output_dir)}")
    print("  Pipeline complete ✓")


# ==============================================================================
if __name__ == "__main__":
    main()
```